James Ridey

Software Engineer | Frontend, Product Engineering & Performance

Sydney, Australia • Australian citizen (E-3 visa eligible) • Seeking relocation to the San Francisco Bay Area
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Software engineer with frontend experience at Nullify, Canva, and Google. Seeking challenging frontend and broader software roles with technical ownership, complex products, and room to learn.

Technical Skills

Professional: TypeScript, JavaScript, React, Go, Python, AWS.

Frontend & tooling: Vite, Playwright, Storybook, ESLint, RAG interfaces, LLM-assisted development.

Projects: C/C++, CUDA, Verilog, Arduino, ESP32, STM32, PCB design.

Professional Experience

Nullify Product Engineer | AI application security — Mar 2026 – present

- Primary owner of UI2: shipped a Vite rebuild with an atomic design architecture spanning approximately **470 components**; owned the architecture and implemented most features.
- Set UI direction, using Claude Design to explore designs. Improved performance, mobile support, visual consistency, navigation flows, and accessibility scores.
- Built opt-in Playwright/Storybook visual regression checks with ImageMagick comparisons; added end-to-end tests for UI flows. Directed LLM implementation of ESLint architecture and design rules.
- Removed Go startup reflection, cutting Lambda initialisation from **1.9 s to 1.1 s** in a development deployment. Reused a Python subprocess across warm invocations, cutting Python startup overhead from **9.7 s to 50 ms**.

Canva Software Engineer — 2020 – 2025

- Integrated Magic Design's AI-generated text and selected photos into preset layouts outside the editor. Reused Canva's canvas rendering with editor-team input to display image previews without each design's full DOM.
- Built Team Brain's frontend with a backend engineer: a working RAG prototype answering questions from documents.

Google Software Engineer Intern — 2017 – 2019; two summer internships

- Ported Google's Neighbourly app to KaiOS/JioPhone with 512 MB RAM; wrote the first porting specification.

ThoughtDesign Junior Software Engineer — 2014 – 2017

Independent Projects & Contributions

OnnxStream & E-paper Art Frame C++ inference & Raspberry Pi

- Added SDXL Turbo support to OnnxStream, adapting tensor dimensions and tiled decoding; fixed tokenisation and VAE scaling bugs. **Five PRs merged upstream**.
- Built a Raspberry Pi e-paper frame that generates images locally every day, integrating inference, a Python service, web controls, and display colour processing.

Minecraft pack.png Seed Search CUDA & GPU profiling

- Used NVIDIA Visual Profiler to find early thread exits in Minecraft's pack.png seed search. Split filtering and tree-position work into separate CUDA kernels: **roughly 2x performance** on my machine. Merged upstream.

JRB8 Computer Digital logic, Verilog & toolchain

- Designed an 8-bit CPU fabricated through Tiny Tapeout; built its instruction set, assembler, language, compiler, and browser-based compile-flash-run tool.

Brother Embroidery Cartridge Hardware & software reverse engineering

- Built a working replacement cartridge through reverse engineering; published findings and code reused by others.

Security Credentials

Flare-On: Completed all challenges in 2022–2025; **157th / 4,100+** entrants in 2025.

Burp Suite Certified Practitioner (2026) • DEF CON Offensive IoT Exploitation course.

Education

Macquarie University BEng (Honours), Software Design — 2016 – 2020